

Metal Surface Defect Detection using Object Detection Models

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Abstract - Metals are widely used in the manufacturing industry to manufacture a variety of products and it is important to visually inspect the end products. Companies incur financial losses and lose customer trust if defective products are delivered by them. Moreover, current inspection processes are highly manual and time-consuming. To overcome these issues, we propose a method to identify the various types of defects like creases, oil stains, etc. as well as locate them accurately. We have created our custom dataset for defect detection on metal surfaces by merging and processing images from two datasets, NEU-DET, and GC10-DET. State-of-the-art object detection models like YOLOv5, YOLOv7, and Detectron2 (Faster RCNN) have been implemented in this paper. We were able to achieve mAP scores of 77.4%, 77.7%, and 43.4% respectively.

Keywords - Surface defect detection, visual inspection, object detection, mAP

I. Introduction

Of all the products manufactured, 68% of those include different types of metals as their primary component. These products contribute to more than 13% of the worldwide GDP. Visual inspection of metal surfaces is an important step in the manufacturing industry as it helps companies to cut down their losses and build customers' trust [11]. However, the current processes are highly manual and time-consuming.

The present visual inspection methods heavily rely on human interaction, which makes the process costly, ineffective, and time-consuming. Moreover, few defective products can go unidentified which leads to companies suffering financial losses and tarnishing their reputation. In this paper, we suggest a method to automate the visual inspection process using several object detection models. The automated process is comparatively faster and more accurate which helps companies curb their financial losses.

We were able to identify 12 types of defects using two datasets, NEU-DET [13][17] and GC10-DET [12][18], mainly 'crazing', 'crease', 'crescent gap', 'inclusion', 'oil spot', 'rolled pit', 'pitted surface',

'punching role', 'silk spot', 'waist folding', 'water spot' and 'welding line'. Figure 1 depicts these defects in the form of images used in the training dataset. We use state-of-the-art object detection models like YOLOv5, YOLOv7, and Detectron2 (Faster RCNN) to identify and locate the defects on a metal surface and were able to achieve mean Average Precision (mAP) scores of 77.4%, 77.7%, and 43.4% respectively.

The proposed solution has been segregated into 4 sections in this paper, Introduction, Literature Review, Methodology and Results.

II. Literature Review

Y. He et al. [1] propose a new architecture combining CNN, MFN, and RPN to detect and localize defects on metal surfaces. Their end-to-end pipeline employs ResNet for feature extraction, MFN to combine features, and RPN to precisely locate the defects. Xufeng Kou et al. [2] achieved mAP scores of 71.3% on GC10-DET and 72.2% on NEU-DET with their proposed YOLOv3 architecture. They were able to reduce computational times by using an anchor-free feature selection method. Xuyi Yu et al. [3] achieved a mAP score of 79.1% on the NEU-DET with their proposed ES-NET architecture. ES-NET [3] contains modified modules like AFGM to combine features and SAE to provide comprehensive guidance for shallow feature map. Yibin Huang et al. [4] introduce MCuePush U-Net to reduce the inference time for surface defect detection from 0.5 seconds to 0.07 seconds. Muhieddine Hatab et al. [5] train YOLOv3 on a dataset obtained from NorthEastern University using 25000 epochs and achieve a mAP score of 70.66%. Chenglong Wang et al. [6] propose a boosting module that consists of a SAF block, CRMA scheme, and PA label smoothing block which was able to achieve a gain of 6.51% and 3.76% over the baseline YOLOv4 and YOLOv5 models.

III. Methodology

A. Dataset

In this paper, we have created our custom dataset by merging two open-source datasets, NEU-DET [13][17] and GC10-DET [12][18]. We have removed ‘patches’ as a class and combined ‘inclusion’, ‘scratches’, and ‘rolled in scale’

classes from NEU-DET [13][17] with ‘inclusion’, ‘creases’, and ‘rolled_pit’ classes respectively from GC10-DET [12][18]. Table 1 shows the number of images in each class in our custom dataset.

Our dataset consists of 3795 images (grayscale images) split into 12 classes, with 80% of the data used for training, 10% for validation, and 10% for testing.

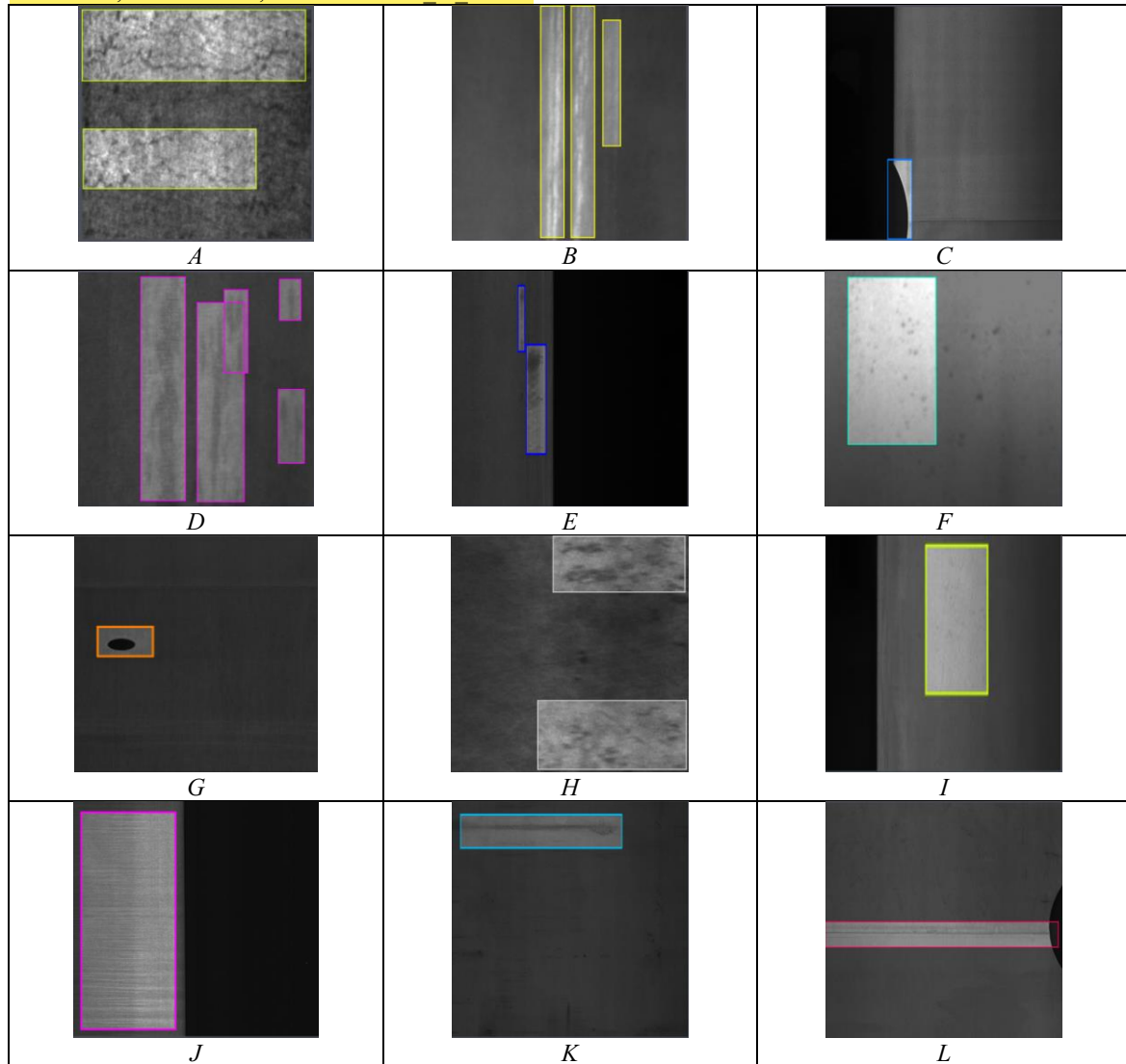


Figure 1 : Custom Dataset, A : Craze, B : Crease, C : Crescent Gap, D : Inclusion, E : Oil Spot, F : Pitted Surface, G : Punching Hole, H : Rolled pit, I : Silk spot, J : Waist folding, K : Water spot, L : Welding Line

Dataset	Class name	No. of Images
1.	Craze	300
2.	Crease	353
3.	Crescent Gap	264
4.	Inclusion	554
5.	Oil spot	250
6.	Pitted Surface	301
7.	Punching Hole	328
8.	Rolled pit	346
9.	Silk spot	734
10.	Waist folding	140
11.	Water spot	309
12.	Welding line	512

Table 1 : Number of Images in each class

B. YOLOv5

In this paper, we have used YOLOv5s (small) and trained it on our custom dataset. The annotation files consist of the class, X coordinate, Y coordinate, and height and width of the bounding box [8][15]. To train the model on our dataset, we change the data-configuration file and provide paths to train, validation, and test datasets [15].

C. YOLOv7

In this paper, we have used YOLOv7 and trained it on our custom dataset. YOLOv7 has been able to reduce 40% of parameters to give low inference time while maintaining high accuracy. YOLOv7 merges multiple computational modules into one at the inference stage, thus giving us low inference time [7][14][16]. It provides model scaling to scale up or down and makes it compatible with any computing device [14][16]. Similar to YOLOv5, it uses the same annotation format and requires similar data-configuration file changes [15].

D. Detectron2

Detectron2 is a library built by researchers at Facebook to support implementations of various

object detection and segmentation algorithms [10]. It enables training at a much faster speed for different algorithms. It supports different algorithms like RetinaNet, Faster RCNN [9], Fast RCNN, Mask RCNN, and many more. In this paper, we have used Faster RCNN.

IV. Results

Our KPI to compare different models is mAP (mean Average Precision) score. We achieved mAP scores of 77.4%, 77.7%, and 43.4% with YOLOv5, YOLOv7, and Detectron2 respectively.

Detectron2 achieved the lowest inference times per iteration which was 0.103171 seconds per iteration.

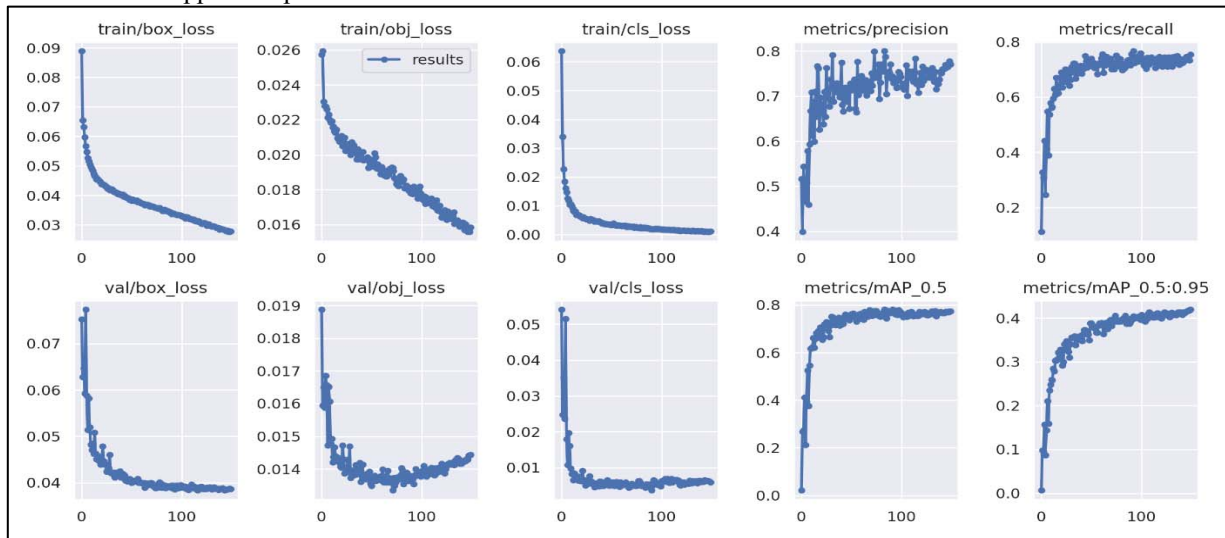


Figure 2 : YOLOv5 Results

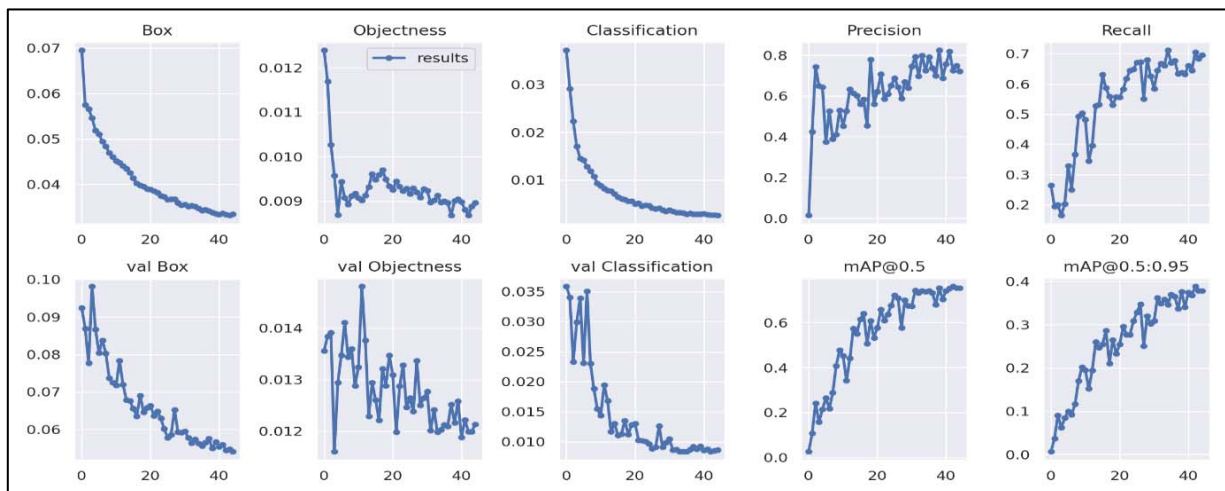


Figure 3 : YOLOv7 Results

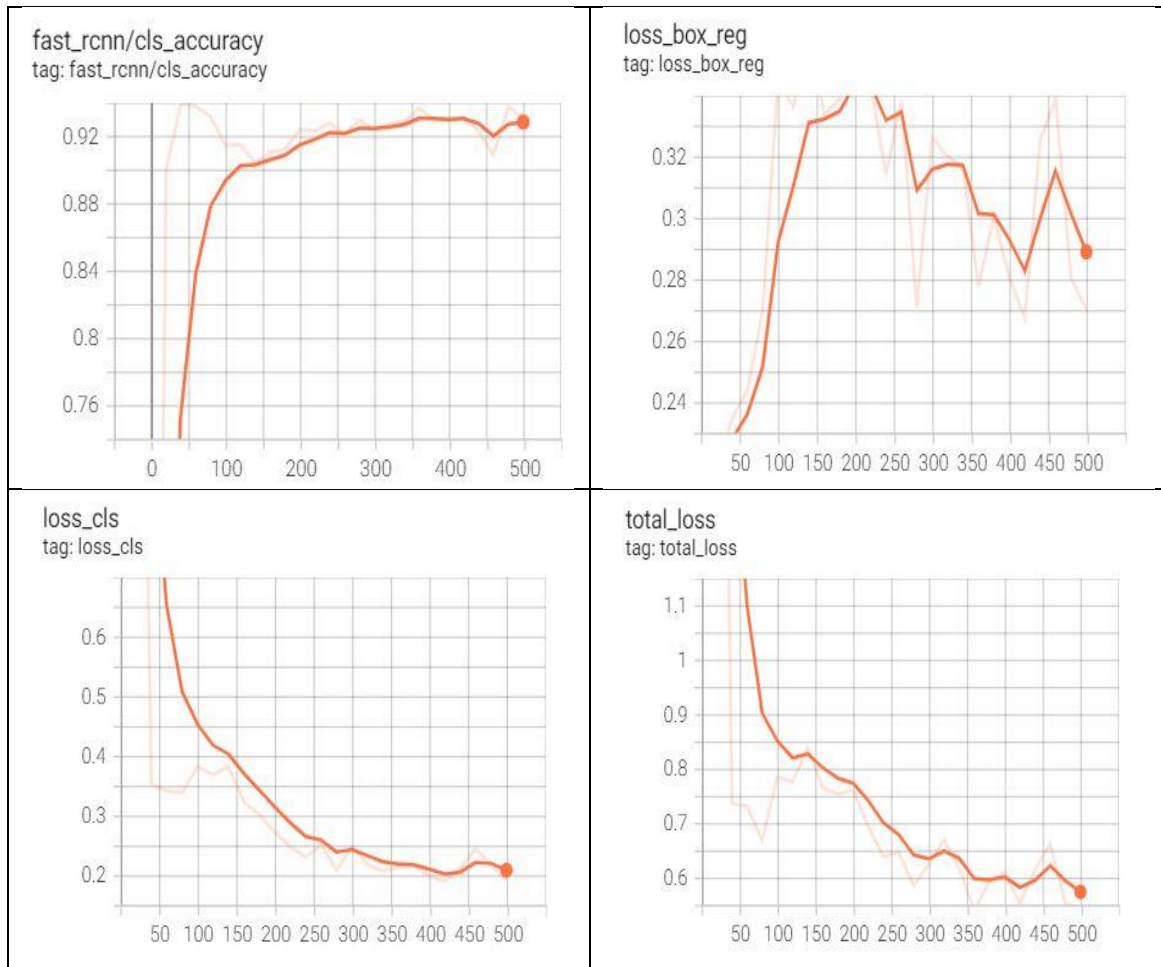


Figure 4 : Detectron2 (Faster RCNN) Results

V. Conclusion

We have trained and tested 3 different object detection models and achieved a mAP score of 77.7% with YOLOv7. Detectron2, on the other hand, gives very low inference times at 103171 sec/iters. This method is helpful to industries to replace the current visual inspection systems which are manual, time-consuming, and prone to errors.

VI. References

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